

Editorial: *Enhancing Reproducibility Through Understanding Datasets*

Public datasets allow published research results to be validated, replicated, and reproduced by the research community, industry, and government agencies. This type of transparency ensures that published findings are trustworthy and accessible as a part of the scientific record. It is important to understand how to use a dataset, and to judge its validity and quality; thus, it becomes necessary to publish articles that specifically focus on the details of the dataset itself. Considering this, I first presented my idea of a “data journal” during the 9 June 2022 IEEE DataPort Steering Committee meeting, and on 7 May 2024, after the hard work of many, we launched IEEE DATA DESCRIPTIONS (IEEE-DATA).

IEEE-DATA is a gold, fully open-access periodical published on behalf of all IEEE societies and technical councils, that publishes peer-reviewed, concise articles detailing datasets across various domains promoting open and reproducible scientific research. The journal’s primary aim is to enhance dataset visibility and utility through comprehensive descriptions, which include data collection methods, data quality assessments, and detailed metadata.

There are three types of articles that can be submitted for review: 1) descriptors; 2) collections; and 3) metadata.

Descriptor articles (the main type of article published) describe the published (or soon-to-be-published) dataset. In essence, this is a detailed manual on the dataset: what it is and how to use it. Similarly, collections articles follow the lead and expectations of descriptor articles, and will focus on groups of datasets collected during or created from a competition, hackathon, or event.

Meta articles depart from but follow the lead and expectations of descriptor articles. Meta articles are articles

that allow for discussion about data. Authors can express their findings about data in several areas, including, but not limited to: standardized formats for data in a given area (e.g., collecting power data from substations), surveys of several datasets of a given type (e.g., GIS and time-series), data that are a library of models for a given AI or machine learning algorithm or system, data or dataset that is metadata of existing published datasets, comparing many similar datasets in an area to create metastatistics data, or any other data topics that may not have traditional scientific experimental results.

IEEE-DATA accepts articles with a recommended length of 4–8 pages. Any extra information that is not inherent to understanding the article content can be submitted as supplementary information and will be published alongside the article on IEEE Xplore. When submitting a paper, authors must choose one of the three article types. Each article type has very specific formatting instructions detailed in the Author Guide. Datasets described in descriptor and collections articles must be stored in a persistent/permanent location (such as IEEE DataPort) and have a DOI. The standard applicable page charge will be heavily discounted until an impact factor is received (usually three years).

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